Synopsis on

AnswerAI

Submitted in Partial fulfillment of the requirements for the degree of Bachelor of Computer Application

CT GROUP OF INSTITUTIONS

Submitted To:

Prof. Joy Karan

Submitted by:

Loveleen Sahota (11792330967)

Vishal Sharma (11792330942)

CT INSTITUTE OF HIGHER STUDIES, JALANDHAR
(Affiliated to Guru Nanak Dev University)

O7 Services is an ISO 9001:2015 Certified company founded in 2015. They specialise in a variety of services including Web Development, Mobile Application Development, AI Bot, Machine learning Algorithm, Data Analytics, Cloud Computing, Cyber Security, Custom Software Development, UI/UX design, Hosting services, Digital Marketing, Registration of Domain Names with various extensions, AMC & MMC Services, Bulk SMS and voice calls. O7 Services offers advanced IT solutions supporting the entire business cycle - from consulting to system development, deployment, quality assurance, and 24x7 support. With over 10 years of experience, the company aims to form long-lasting strategic partnerships with clients, offering affordable prices, timely delivery, and measurable business results. Their headquarters are located in Jalandhar with a branch office in Hoshiarpur.

Some of the products developed by O7 Services include Vehicle Tracking System, Invoice Software, School Management System, Hospital Management System, Parents-Teacher Communication App, Fee Management system, Task Management System, Online Food Ordering App, Security App, Admission system, Inventory Software, and Car Servicing App. Additionally, O7 Services provides training programs including 6 Weeks/6 Months Industrial Training, Project-Based Training, Corporate Training, and Job-Oriented Courses Training, covering major IT trends such as Full Stack Development (MEAN/MERN), Flutter, Kotlin Android, Swift UI (iOS), Firebase, Python, Angular, React Js, Vue Js, Node Js, ASP.NET, .NET Core, PHP, Laravel, CodeIgniter, Software Testing, Cloud Computing, Blockchain, DevOps, Data Science, Artificial Intelligence, Machine Learning, UI/UX Designing, Digital Marketing, WordPress, Linux, CCNP, CCNA Security, Network Security, Cyber Security, MCSE, MCITP, Java, Spring, Hibernate, C/C++, Photoshop, Adobe Illustrator, Figma, CorelDraw and many more.

+91- 8437365007, +91-181-5015007

E-Mail: enquiry@o7services.com , hr@o7services.com

www.o7services.com

Chapter 1 Introduction

1.1 Purpose

The purpose of the **AnswerAI** project is to develop an industry-ready, multilingual document intelligence system that enables users to extract accurate and context-aware answers from uploaded documents using deep learning models and Retrieval-Augmented Generation (RAG). The system aims to replace traditional keyword-based document search with a semantic, AI-driven approach that understands user queries, retrieves relevant information through vector-based similarity search, and generates precise responses grounded in stored documents.

By supporting multiple file formats and languages, the project enhances accessibility, reduces manual effort in information retrieval, minimizes incorrect or fabricated responses, and provides an efficient, scalable solution suitable for educational, corporate and enterprise environments.

The system aims to bridge the gap between traditional document search and intelligent AI-powered knowledge retrieval systems.

1.2 Scope of the Project

- Can be used in educational institutions
- Useful for legal document search
- Corporate policy analysis
- Research document summarization
- Enterprise knowledge management

1.3 Architecture Diagram

User → Upload Document → Text Extraction → Embeddings → FAISS → Query → RAG Model → Answer with Source

1.4 Keywords

- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning
- Natural Language Processing (NLP)
- Retrieval-Augmented Generation (RAG)
- Document Question Answering
- Multilingual Chatbot
- Semantic Search
- Transformer Models
- Vector Embeddings
- FAISS (Similarity Search)
- Large Language Models (LLMs)
- Knowledge Retrieval System
- Intelligent Document Processing (IDP)

Chapter 2

Existing and Proposed System

2.1 EXISTING SYSTEM

The existing document search and question-answering systems primarily rely on keyword-based search mechanisms, where users must manually scan through large documents to find relevant information. These systems lack semantic understanding and often return irrelevant or incomplete results if the exact keywords are not matched. Traditional chatbots, on the other hand, generate responses based only on pre-trained knowledge and do not retrieve information directly from user-uploaded documents, which can lead to inaccurate or hallucinated answers. Additionally, most legacy systems provide limited support for multiple document formats and multilingual queries, making them inefficient for modern organizational and enterprise-level document management needs.

2.2 PROPOSED SYSTEM -

- The proposed system uses **Retrieval-Augmented Generation (RAG)** to combine semantic document retrieval with deep learning–based answer generation, ensuring accurate and context-aware responses.
- It supports **multiple document formats** such as PDF, DOCX and TXT, allowing users to upload and query various types of files within a single platform.
- The system enables **multilingual question answering**, allowing users to ask queries in different languages and receive responses accordingly.
- It provides **source-referenced answers**, displaying document names and relevant sections to ensure transparency, reliability, and reduced hallucination.

Chapter 3

Disadvantages of Existing System

3.1 Disadvantages of Existing System -

- Time-consuming manual document searching.
- Keyword-based search lacks semantic understanding.
- High chances of irrelevant or incomplete results.
- Traditional chatbots may generate incorrect or hallucinated answers.
- Limited support for multiple file formats.
- Lack of multilingual query support.
- No proper source referencing for generated answers.

Chapter 4

Features of Project

- Multi-format document support (PDF, DOCX, TXT).
- Semantic vector-based document indexing using embeddings.
- Retrieval-Augmented Generation (RAG) architecture for accurate answers.
- Transformer-based deep learning models for context-aware response generation.
- Multilingual question and answer support.
- Source citation with document name and relevant section/page reference.
- Secure document storage with user-based access control.
- Fast similarity search using vector database (FAISS or equivalent).
- Scalable and modular system architecture.
- User-friendly web interface for document upload and querying.

Chapter 5

Technology used under my project

1. Programming Language

Python

● Core programming language used for backend development.

● Handles document processing, AI model integration, and API development.

● Supports advanced libraries for AI, ML, and NLP.

2. Deep Learning & NLP

Transformer Models (BERT / T5 / LLaMA-based models or more)

● Used for context-aware answer generation.

● Enables natural language understanding and response creation.

Sentence-Transformers (SBERT or more)

● Generates semantic embeddings for documents and queries.

● Improves similarity-based retrieval accuracy.

3. Retrieval System

FAISS (Facebook AI Similarity Search)

● Used for efficient vector similarity search.

● Stores and retrieves document embeddings quickly.

BM25 (Optional Hybrid Search)

● Enhances retrieval by combining keyword and semantic search.

4. Document Processing

PyMuPDF / pdfplumber

● Extracts text from PDF documents.

python-docx

● Reads content from DOCX files.

NLTK / spaCy

● Performs text cleaning, tokenization, and preprocessing.

5. Web Framework

Flask / FastAPI / Streamlit

● Used to build backend APIs and user interface.

● Handles file uploads, user queries, and response display.

6. Database & Storage

Vector Database (FAISS / Pinecone)

● Stores document embeddings for semantic search.

Relational Database (MySQL / SQLite)

● Stores user data, document metadata, and logs.

7. Translation & Multilingual Support

MarianMT / mBART Models

● Enables multilingual query processing and response generation.

Language Detection Libraries (langdetect / fastText)

● Detects the language of user queries automatically.

8. Development & Deployment Tools

Git & GitHub

● Version control and project collaboration.

Docker (Optional)

● Containerization for easy deployment.

VS Code / Jupyter Notebook

● Development and experimentation environment.

Chapter 6

Software and hardware Requirements

6.1 Software Requirements-

Front End Tool	:	Streamlit / React JS (for User Interface)
Back End Tool	:	Python 3.8 or above (Flask / FastAPI)
Deep Learning Framework	:	PyTorch / TensorFlow
NLP & Embedding Libraries	:	Hugging Face Transformers, Sentence-Transformers
Vector Database	:	FAISS (Facebook AI Similarity Search)
Database	:	MySQL / SQLite
Browser	:	Google Chrome / Microsoft Edge
Operating System	:	Windows Operating System / Linux
Web Server	:	Flask Development Server / Uvicorn / Apache

6.2 Hardware Requirements-

Processor	:	i5 recommended
RAM	:	8.00 GB (16 GB recommended for better performance)
Hard Disk	:	100 GB or higher
GPU (Optional)	:	NVIDIA GPU for faster model processing
Internet Connection	:	Required for model downloads and updates